DATA ANALYSIS REPORT

PREDICTING WATER WELL FUNCTIONALITY FOR TANZANIA

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Date: *September 12th, 2025*

Overview

Access to safe and reliable water remains a daily uncertainty for millions of Tanzanians. In 2025, Tanzania's population is estimated at 70.5 million, with nearly one in three people, around 24–25 million, relying on wells and boreholes as their primary drinking water source. These wells are more than infrastructure; they are lifelines supporting health, livelihoods, and dignity. Yet, many fail within just a few years due to poor construction, environmental stress, limited maintenance, or lack of spare parts.

The consequences of well failures are profound: communities are exposed to waterborne diseases, households incur economic losses securing alternative sources, and the burden of water collection disproportionately affects women and children, limiting educational opportunities and economic participation. Additionally, donor and government investments are wasted when wells collapse prematurely.

This challenge is not unique to Tanzania, but the scale is striking. While expanding water access remains important, sustainability; keeping existing water points functional is equally critical. The Water Sector Development Program (WSDP) recognizes that predictive insights into well functionality are key to achieving Sustainable Development Goal 6: Clean Water and Sanitation.

This project frames well functionality as a data science problem, aiming to predict whether a well is functional, needs repair, or is non-functional. Such predictive modeling offers multiple benefits:

- Proactive maintenance: Technical teams can prioritize wells at risk of failure, preventing downtime.
- Smarter investments: Resources can be directed to areas where wells are most vulnerable.
- Community resilience: Reliable access improves health, education, and economic participation.
- Accountability and transparency: Data-driven decision-making builds trust between communities, government, and development partners.

Ultimately, this project goes beyond technical analysis. It seeks to shift the focus from merely building wells to building resilience, ensuring that every investment in water infrastructure delivers lasting impact for Tanzanian communities.

Problem Statement

Despite substantial investments in rural water infrastructure, a large proportion of Tanzania's wells remain non-functional or in need of repair at any given time. This unreliability imposes significant health, economic, and social burdens on the millions of people who depend on these water sources. Existing monitoring methods are often reactive, resource-intensive, and slow, leaving communities without dependable access to water for extended periods.

The core challenge is to develop a predictive model capable of classifying wells into three operational categories: functional, functional but needs repair, or non-functional. By proactively identifying wells at risk of failure, policymakers, technical teams, and development partners can prioritize maintenance efforts, minimize downtime, and enhance the sustainability and reliability of rural water access.

This task is approached as a multi-class classification problem, leveraging historical and environmental data; such as well location, construction type, usage patterns, and water quality, to accurately predict the operational status of wells and support data-driven decision-making.

Business Objectives

- **Optimize resource allocation:** Enable water authorities, NGOs, and maintenance teams to prioritize interventions on wells most at risk, reducing downtime and operational costs.
- **Increase investment efficiency:** Ensure funds for drilling, repair, and maintenance deliver maximum long-term impact by focusing on wells that need attention the most.
- **Enhance community impact:** Improve access to reliable water, thereby reducing health risks, minimizing economic loss, and alleviating the burden on women and children who fetch water.

- **Support data-driven planning:** Provide policymakers and development partners with actionable insights to guide infrastructure development, maintenance schedules, and future data collection strategies.
- **Promote transparency and accountability:** Track well performance over time and communicate results to stakeholders, reinforcing trust between communities, government agencies, and donors.
- **Scalability and adaptability:** Develop a framework that can be replicated across other regions or countries facing similar rural water challenges.

Data Source

This project uses the Tanzania Water Wells dataset, compiled by Taarifa in collaboration with the Tanzanian Ministry of Water. The dataset has been made publicly available through the DrivenData “Pump It Up” competition.

It contains detailed records of rural water points across Tanzania, including attributes such as well location, construction type, water quality, management practices, and operational status. This comprehensive dataset provides a rich foundation for analyzing well functionality and developing predictive models to support proactive maintenance and sustainable water access initiatives.

Data Constraints

- **Temporal drift:** Wells may have degraded or management practices may have changed since the data was collected, meaning patterns learned may not fully reflect current conditions.
- **Historical relevance:** Predictions for today’s wells may be less accurate if based solely on older data.
- **Model generalization:** The model may capture trends specific to the historical period, limiting its applicability to current or future scenarios.
- **Data quality:** The data has missing values, inconsistent string entries and redundant repetitive columns that were tedious to clean.
- **Geographic scope:** The dataset covers Tanzania only; applying the model elsewhere would require local retraining and adaptation.
- **Dynamic factors:** Sudden environmental shocks (e.g droughts, floods, contamination) are not fully captured by static historical data.

Data Understanding

Dataset Overview

The Tanzania Water Wells dataset provides detailed records of rural water points across the country, capturing information about their location, construction, management, operational status, and user practices. It has 59,400 rows and 41 columns in total.
The data was provided in 3 separate CSV (Comma Separated Values) files:

1. **Training set values.csv -** containing the features for target prediction; otherwise known as the independent variables.
2. **Training set label.csv -** containing the target variable; otherwise known as the dependent variable.
3. **Test set values.csv -** containing test features.

Key attributes include:

- Target variable:
  - status_group: Operational status of the well, with three categories:
    - functional
    - functional needs repair
    - non-functional

- Well specifications and capacity:
  - amount_tsh: Total static head, representing the amount of water available at the well, recorded in meters.
  - construction_year: Year the well was constructed
  - waterpoint_type, waterpoint_type_group: Type of waterpoint
  - extraction_type, extraction_type_group, extraction_type_class: Type of water extraction

- Location and geography:
  - gps_height, longitude, latitude: Geographic coordinates and altitude
  - basin, subvillage, region, region_code, district_code, lga, ward: Various geographic identifiers

- Management and governance:
  - funder, installer: Organizations responsible for funding or installing the well
  - scheme_management, scheme_name, management, management_group: Operational management details
  - permit, public_meeting: Regulatory and community engagement indicators

- Water characteristics:
  - water_quality, quality_group: Water quality

    - quantity, quantity_group: Water quantity
    - source, source_type, source_class: Water source

- Financial aspects / user payment:
    - payment: What users pay for water (numeric or categorical cost)
    - payment_type: How users pay for water.

- Population and record metadata:
    - population: Number of people relying on the well
    - recorded_by, date_recorded, wpt_name, num_private: Metadata about data entry and identification

This dataset provides a rich foundation for analyzing drivers of well functionality, building predictive models, and generating actionable insights for maintenance prioritization, policy planning, and resource allocation.

Data Preparation

1. Merging Training Features with the Target

- To facilitate comprehensive Exploratory Data Analysis (EDA) and modeling, the training features were merged with the target variable, resulting in a complete dataset ready for analysis.

2. Dropping Unnecessary Columns

- Several columns were redundant or offered minimal analytical value.
- For instance, scheme_management, management, and management_group all captured similar information regarding the administrative bodies overseeing water wells. Columns like id and num_private provided little predictive power.
- After careful evaluation, these columns were removed, reducing the dataset from 41 columns to 18, streamlining subsequent analysis.

3. Handling Missing Data

- Only two columns contained missing values: installer and permit.
- Nulls were dropped to ensure data completeness and integrity.

4. Data Type Conversion

- Date columns were converted to datetime format to extract year-related information, allowing for insights into the age and temporal trends of water wells.

5. Handling Duplicates

- The dataset was verified to be free of duplicate records, ensuring data integrity.

6. Handling Outliers

- Outliers were addressed with care.
- Genuine extreme values, such as wells located below sea level (gps_height), were retained, while obvious data entry errors were identified, converted to nulls, and subsequently filled with placeholder values.

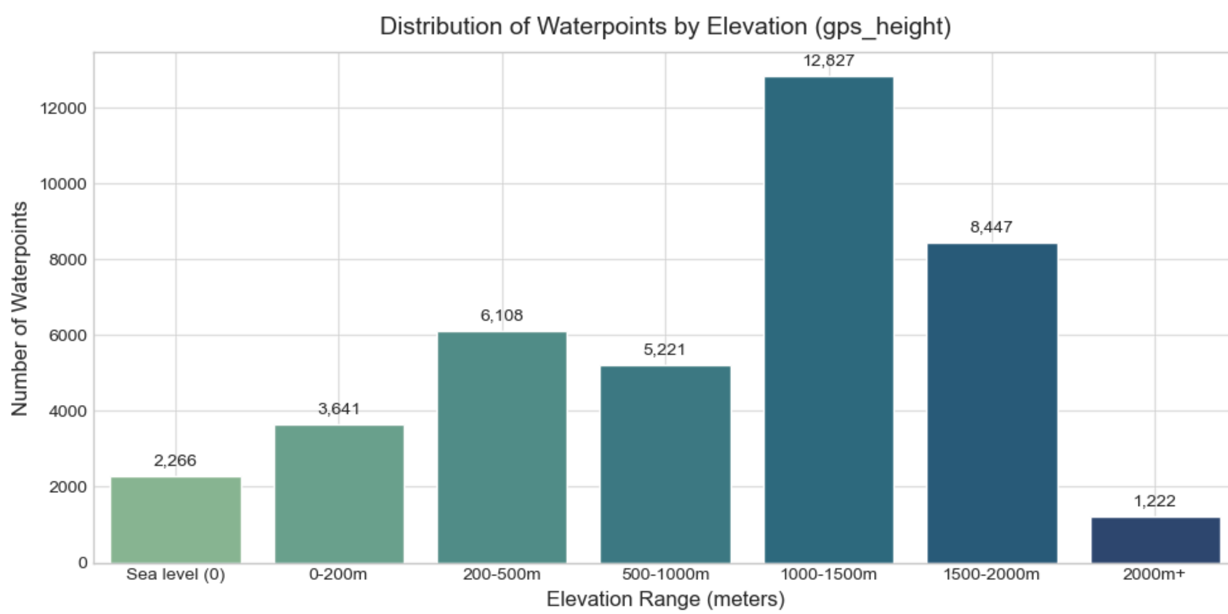
7. Feature Engineering

- Categorical Variables: One-hot encoding was applied to prepare them for machine learning models.
- Target Variable: Label encoding was performed for compatibility with predictive algorithms.
- Numerical Features: Log transformation and scaling were applied to normalize distributions and ensure consistent feature ranges for modeling.

Exploratory Data Analysis

Univariate Analysis

1. Waterpoints by elevation (gps_height)



The figure above shows the distribution of water points by elevation

Observation:

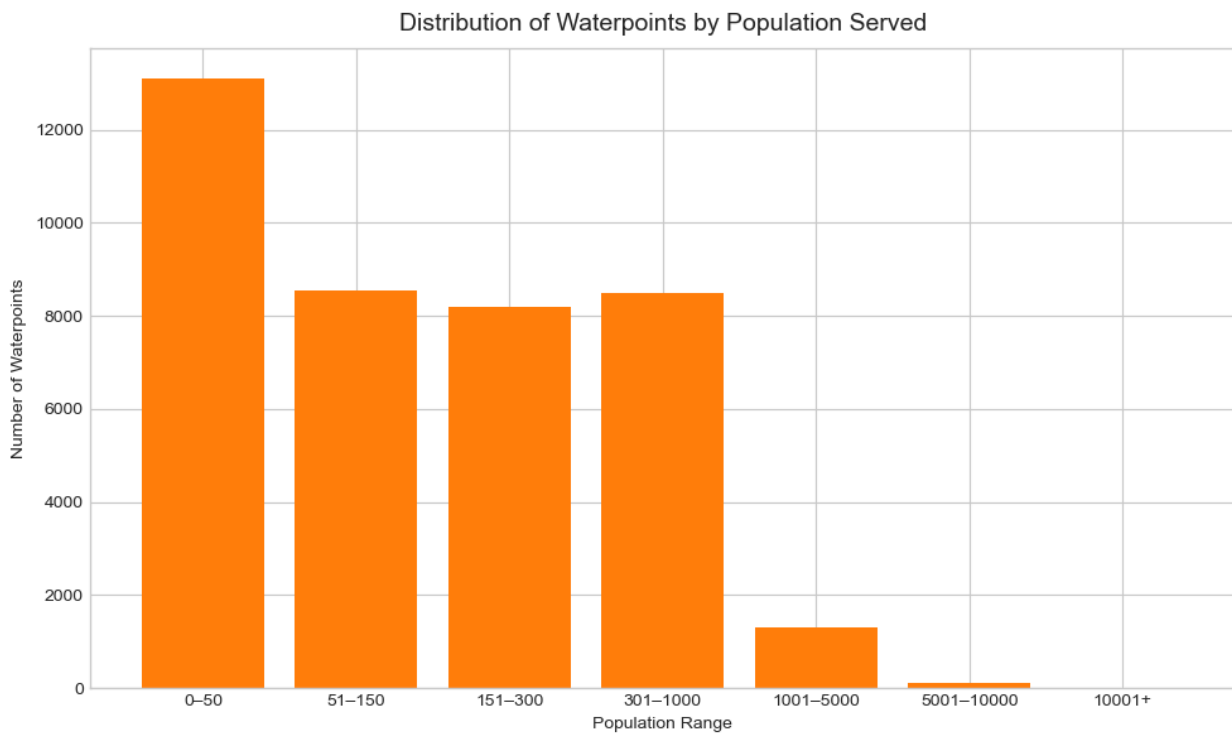
- The distribution of waterpoints by elevation shows that the majority of wells are situated between 1,000–1,500 meters (12,827 wells) and 1,500–2,000 meters (8,447 wells).

- Fewer wells are found at extreme elevations, with sea level locations hosting 2,266 wells and elevations above 2,000 meters hosting only 1,222 wells.

Insight:

- This indicates a tendency for waterpoints to be concentrated in mid-altitude regions, likely reflecting population density and accessibility factors.

2. Population

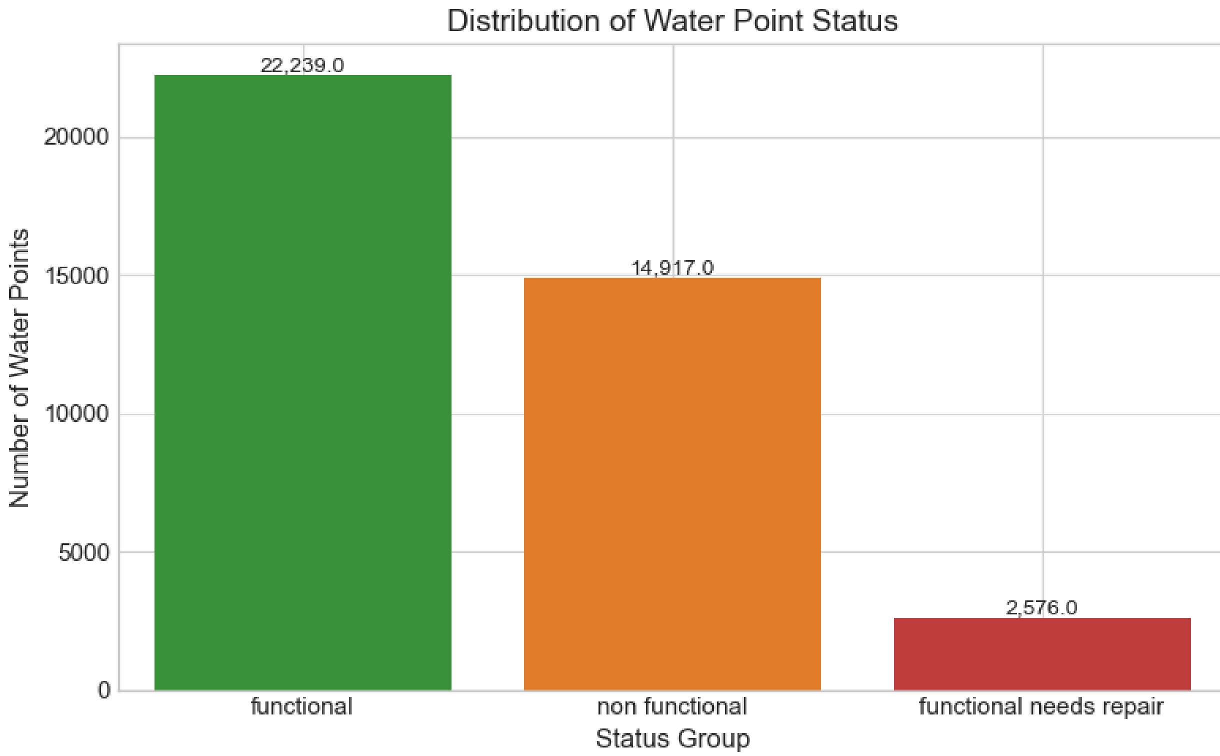


The figure above shows the distribution of waterpoints by population served

Observation:

- Approximately 75% of waterpoints serve 300 people or fewer. Values range from 0, likely representing abandoned or unused waterpoints, up to 30,500, which correspond to large peri-urban or urban settlements.
- Larger populations place greater demand on waterpoints, increasing the likelihood of breakdown or functional stress.
- Conversely, zero-population waterpoints may reflect disuse.

3. Waterpoint Status

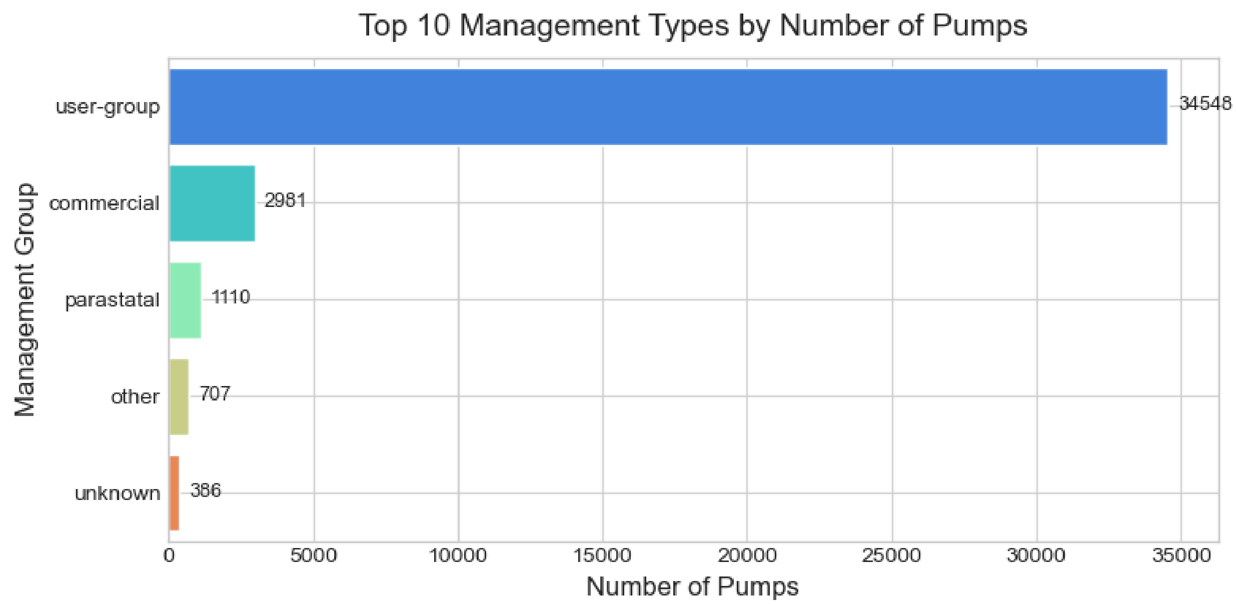


The figure above shows the distribution of waterpoints by functionality status

Observation:

- Most water points in the dataset are fully operational, highlighting that basic water access is largely maintained.
- However, a substantial number; nearly 15,000, are non-functional, pointing to persistent maintenance and infrastructure challenges.
- A smaller segment requires repairs despite being functional, signaling opportunities for preventive maintenance to sustain service reliability.

4. Management



The figure above shows the distribution of waterpoint management entities

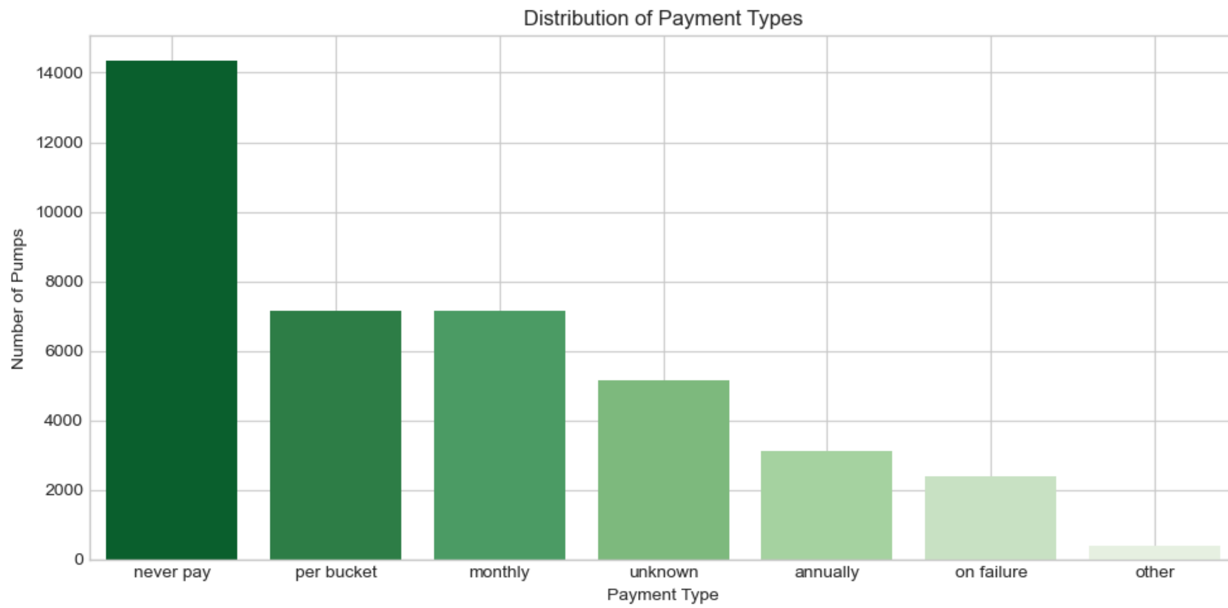
Observation:

- The majority of water points (32,579) are managed by user groups, emphasizing the central role of community-based management.
- Commercial entities oversee a smaller portion (2,644), while parastatals manage 934 water points.
- A minor number of water points fall under other (507) or unknown (338) management types.

Insight:

- Overall, the distribution is heavily skewed toward community-driven oversight.

5. Payment Type



The figure above shows the distribution of how users contribute financially for water services

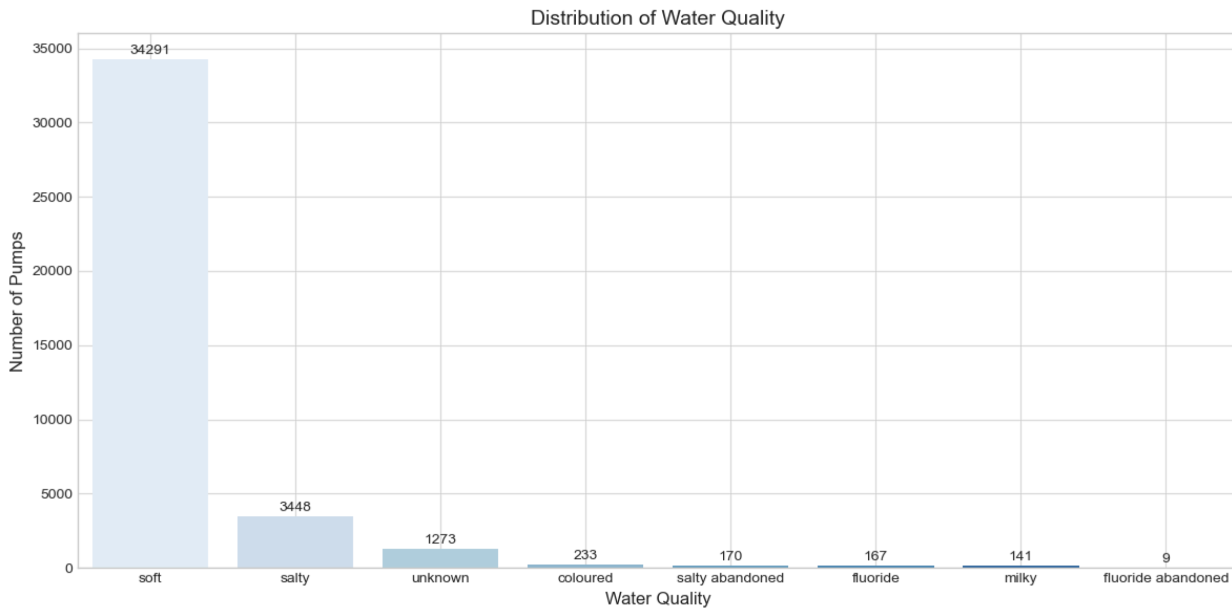
Observation:

- Most water points operate without direct user payments, with “Never pay” accounting for 14,340 points.
- Usage-based or regular contributions; like per bucket (7,162) and monthly (7,161), are also common.
- Some water points (5,169) have unknown payment status, revealing either these are free communal wells or gaps in the data.
- Less common schemes include annual payments, payment on failure, or other arrangements.

Insight:

- The pattern suggests that many water points rely on non-paying or small-scale contributions, which could impact maintenance and long-term sustainability.

6. Water Quality



The figure above captures the chemical and physical characteristics of water points

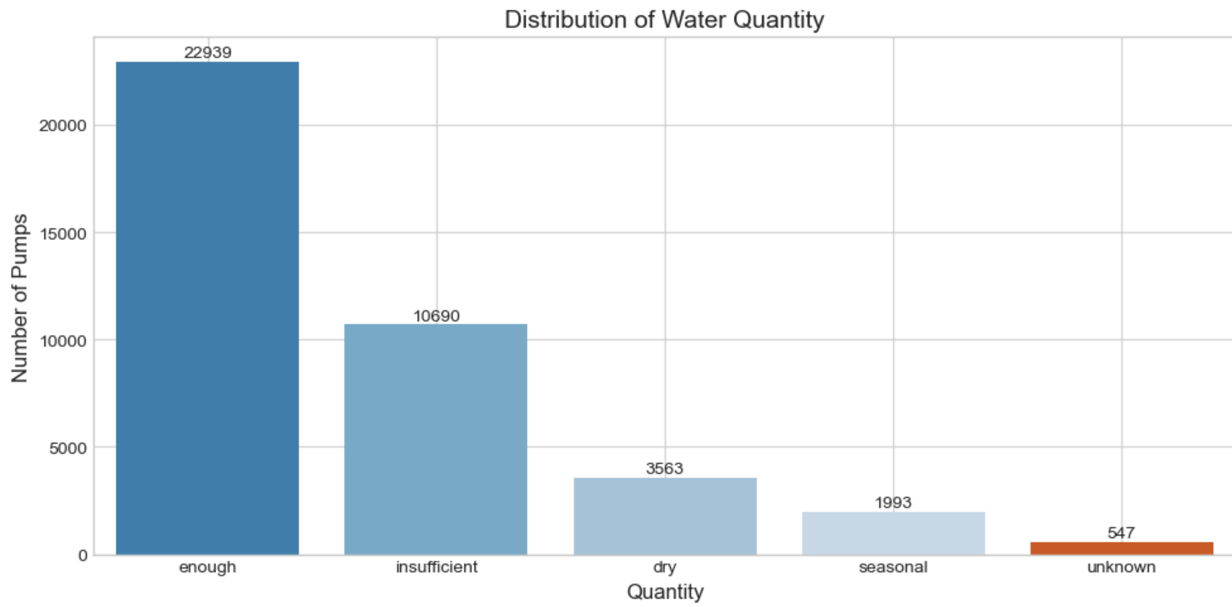
Observation:

- The majority of water points (32,456) provide soft water, indicating generally good water quality across the country.
- A smaller portion is salty (3,157), while other quality types; coloured (233), fluoride (165), and milky (137), are relatively rare.
- A few points are marked as abandoned due to salinity or fluoride issues (75 in total), and 779 water points have unknown water quality.

Insight:

- Soft water dominates, but pockets of poor or uncertain water quality highlight areas that may need monitoring or intervention.

7. Quantity



The figure above represents the availability of water at each point

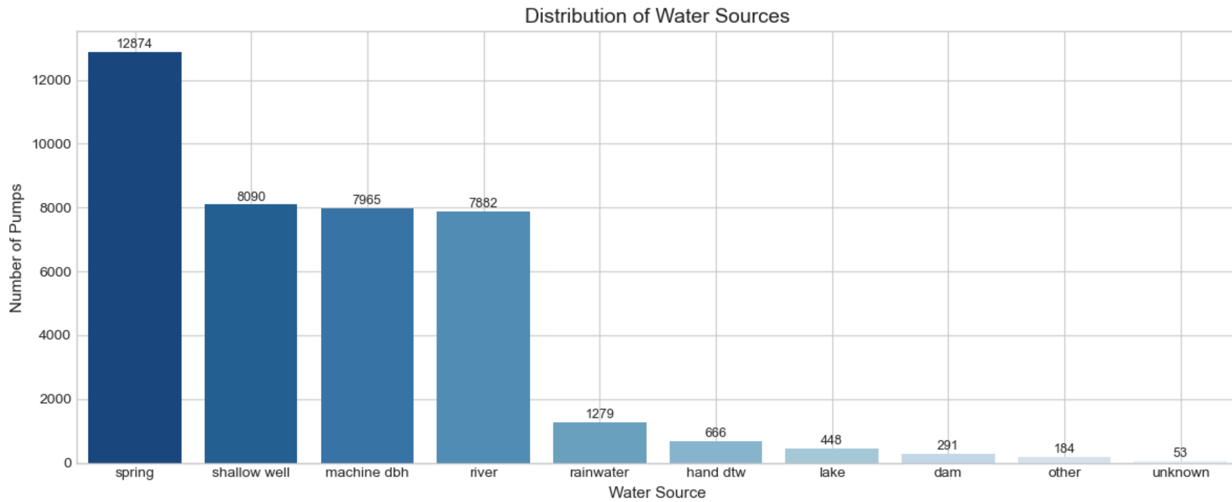
Observation:

- Most water points (21,937) provide an adequate water supply, indicating that a majority reliably meet community needs.
- However, a significant number have insufficient water (9,785), and smaller portions are dry (3,022) or seasonal (1,822), reflecting variability in availability.
- A few water points (436) have unknown quantity status.

Insight:

- While many water points supply enough water, a substantial minority face challenges in consistent availability, which could affect community access and planning.

8. Water Source



The figure above represents the origin of water at each point

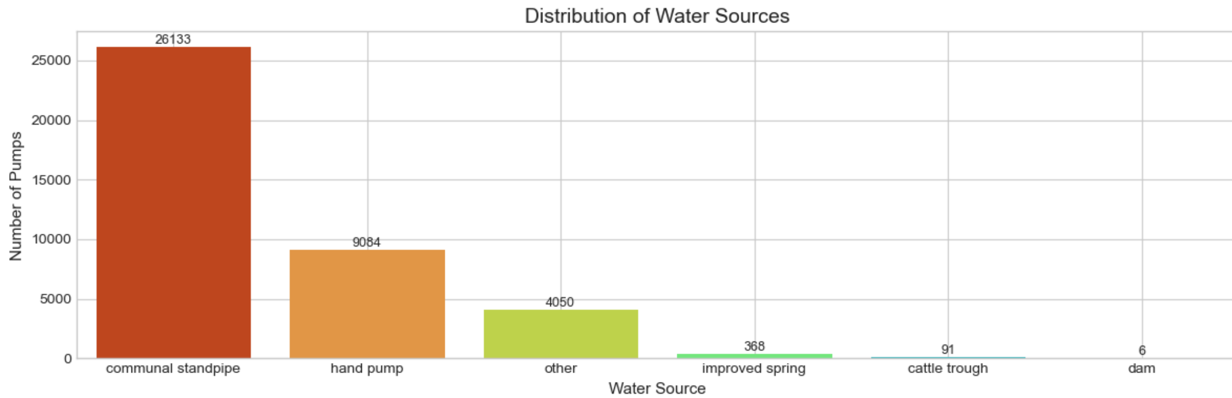
Observation:

- Springs are the most common water source, serving 12,623 water points, followed by rivers (7,754) and shallow wells (7,433).
- Machine-drilled boreholes (6,592) also contribute significantly.
- Less common sources include rainwater (1,030), hand-dug wells (631), lakes (448), and dams (265).
- A small number of water points fall under other (181) or unknown (45) categories.

Insight:

- Natural surface and groundwater sources dominate, with engineered solutions playing an important but smaller role.

9. Waterpoint Type



The figure above represents the type of infrastructure used for water point access

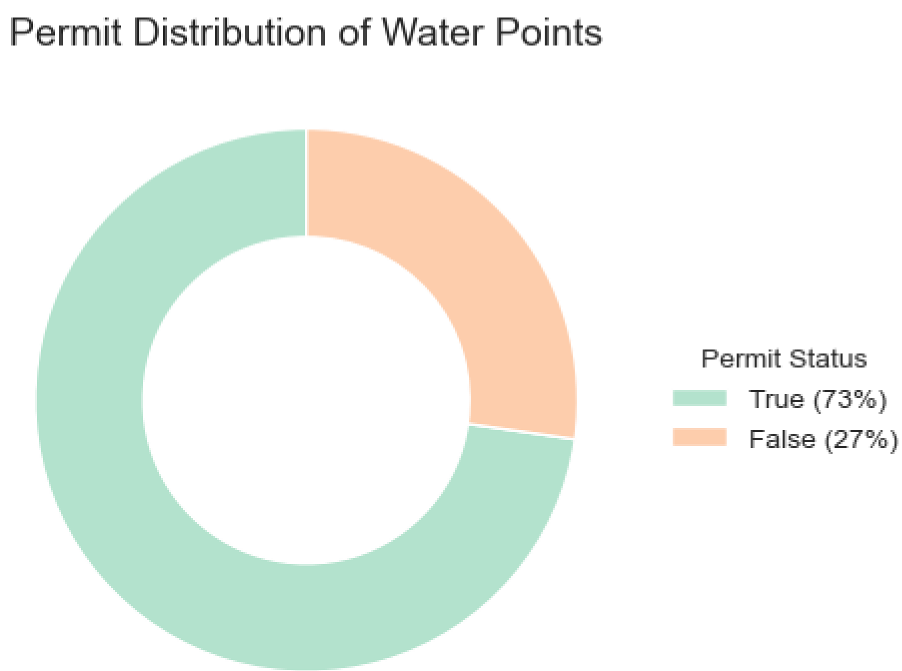
Observation:

- The majority of water points are communal standpipes (24,449), highlighting the prevalence of shared water access points.
- Hand pumps (8,517) are the next most common, followed by a smaller number classified as other types (3,605).
- Improved springs (364), cattle troughs (62), and dams (5) are rare.

Insight:

- The distribution emphasizes reliance on communal infrastructure, with specialized or livestock-focused water points forming a very small fraction.

10.Permit



The figure above represents the regulatory compliance of water points

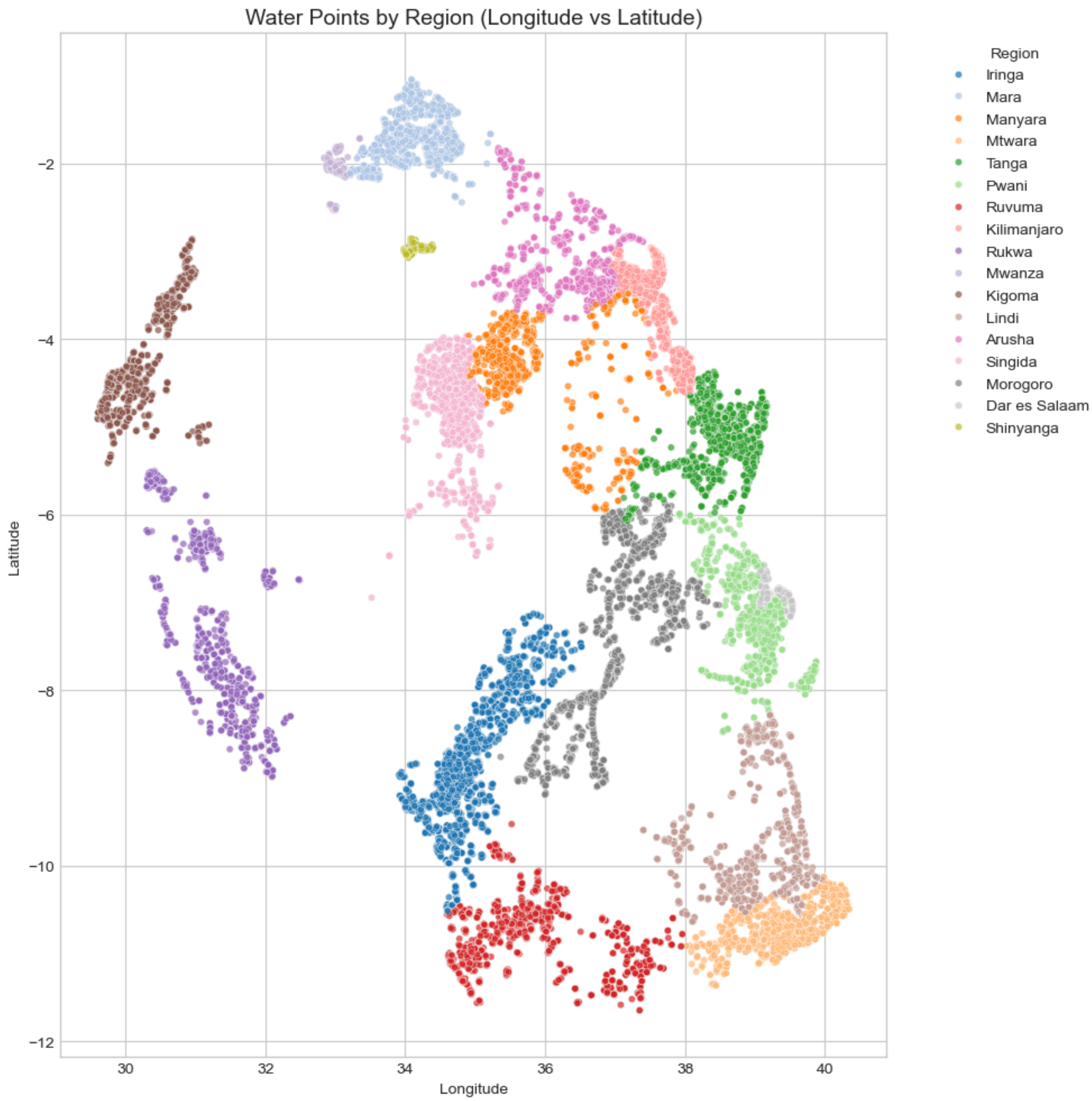
Observation:

- Most water points (26,909) have permits, indicating formal authorization and likely regulatory compliance.
- However, a notable portion (10,093) operate without permits, which may reflect informal setups or gaps in oversight.

Insight:

- While the majority of water points are officially sanctioned, a significant minority lack formal approval.

11.Region



The figure above represents the geographic spread of water points across Tanzania

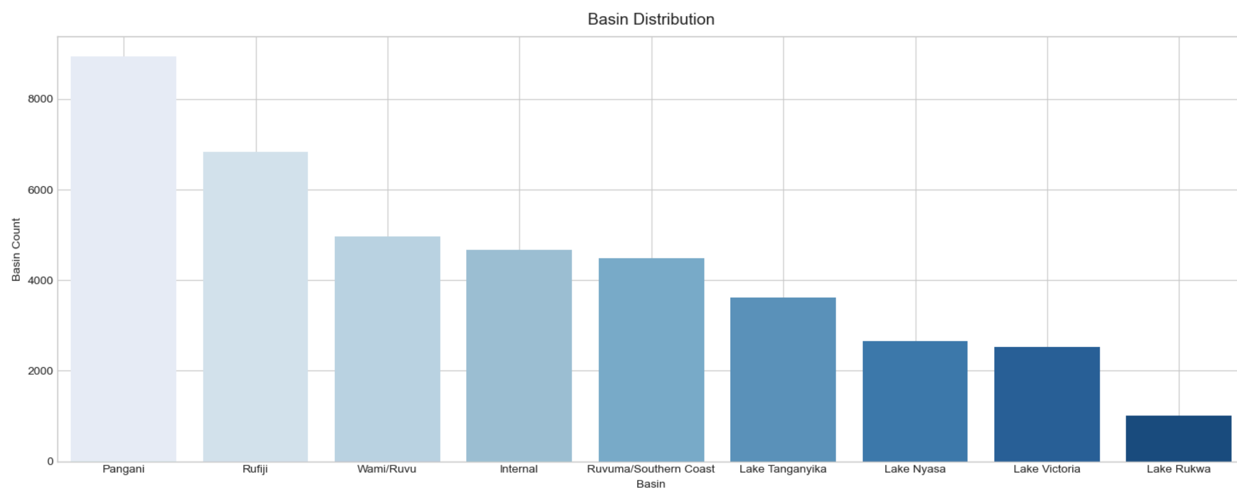
Observation:

- Water points are unevenly distributed across regions. Iringa has the highest number (5,291), followed by Kilimanjaro (4,241) and Morogoro (4,006).
- Regions like Mwanza (367) and Shinyanga (164) have far fewer water points, highlighting significant regional disparities.

Insight:

- The distribution suggests that some regions have far greater access to water infrastructure, which could inform resource allocation and development planning.

12.Basin



The figure above represents the major river basins and lakes across Tanzania

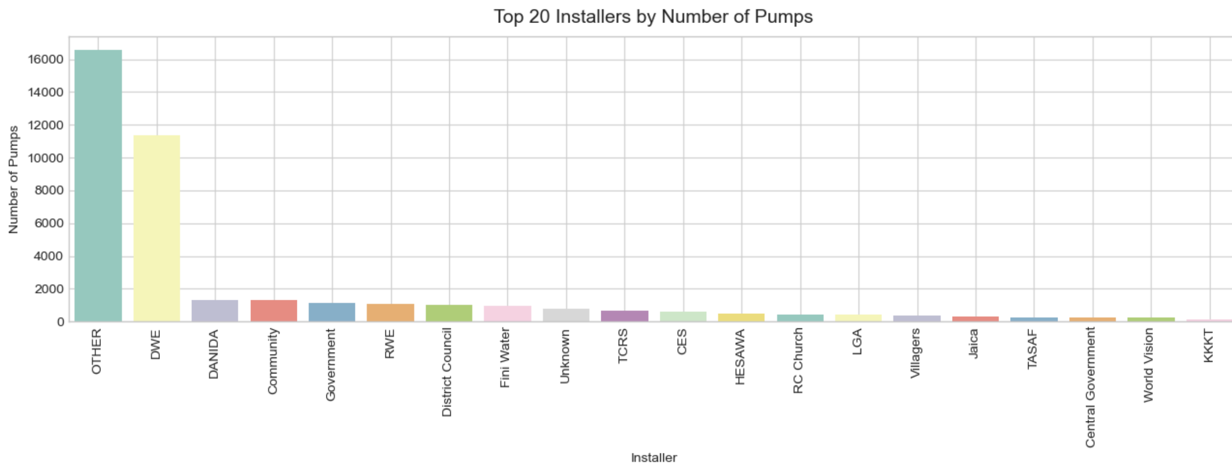
Observation:

- The majority of water points are located in the Pangani (8,678) and Rufiji (6,820) basins, followed by Ruvuma/Southern Coast (4,492) and Wami/Ruvu (4,167).
- Fewer water points are found in basins like Lake Nyasa (2,654), Lake Victoria (2,165), and Lake Rukwa (1,012).

Insight:

- Water point distribution is concentrated in a few major basins, suggesting that water infrastructure planning may need to consider basin-specific needs and resource availability.

13.Installer



The figure above shows the distribution of the top 20 entities responsible for water pump installations

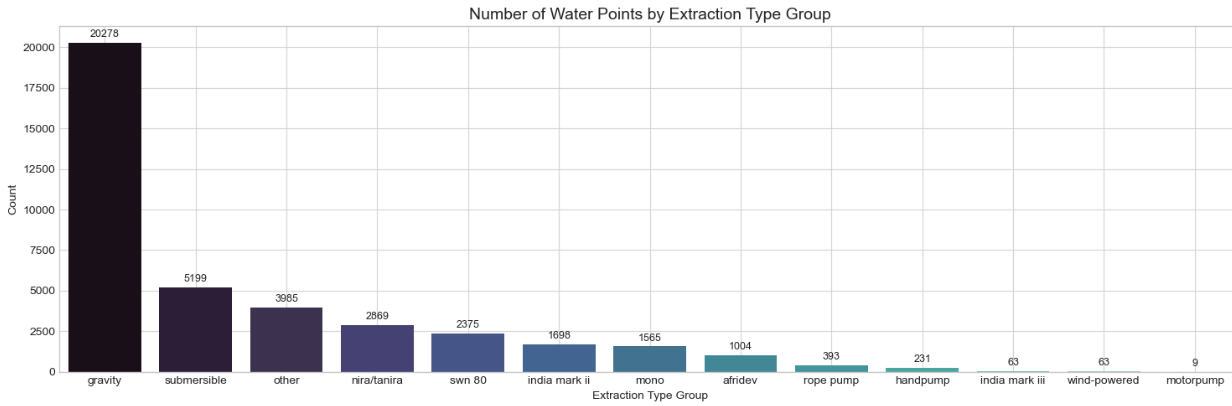
Observation:

- The majority of water points were installed by a few key players.
- “OTHER”; which is a group of all entities that have installed less than 350 wells, accounts for the largest share (15,206), followed by DWE (11,301).
- Other notable contributors include DANIDA (1,331), Community initiatives (1,327), and Government agencies (1,128).
- Smaller contributions come from organizations like RWE, District Councils, and NGOs.

Insight:

- Installation efforts are concentrated among a few major actors, while numerous smaller organizations and community groups handle a minority of water points.

14.Extraction Type



The figure above represents the technology used to extract water from wells

Observation:

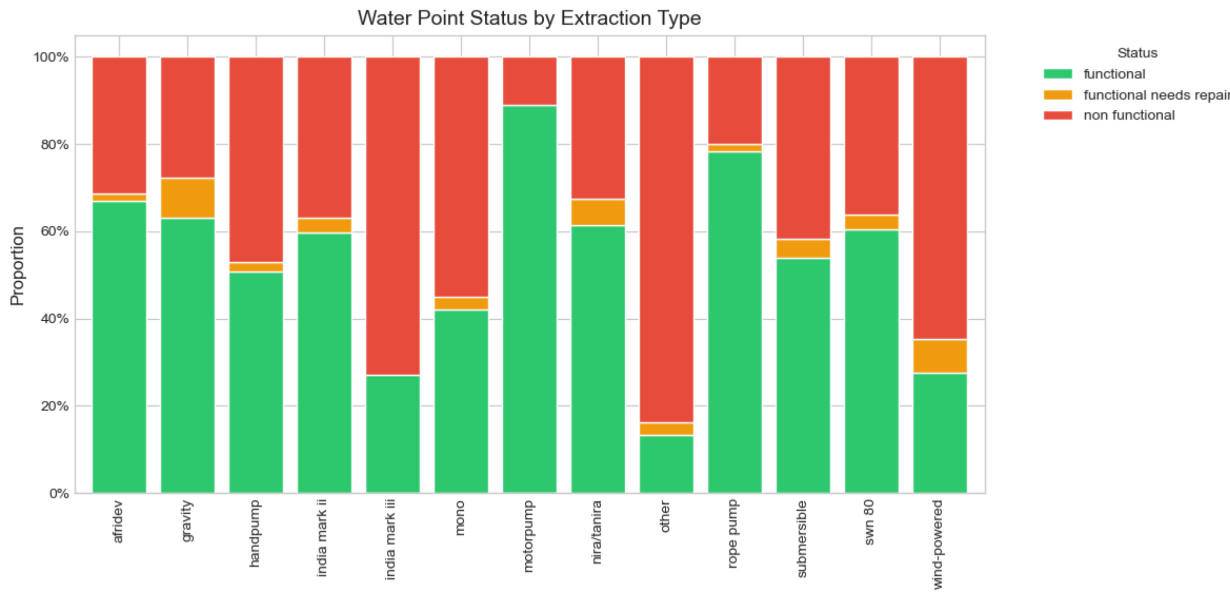
- Gravity-based systems dominate water extraction, serving 19,805 water points, highlighting reliance on natural flow for water delivery.
- Submersible pumps (4,341) and other technologies (3,565) also contribute significantly.
- Less common technologies include Nira/Tanira (2,607), SWN 80 (2,356), India Mark II (1,639), and Mono pumps (1,167).
- Specialized or small-scale solutions like Afridev, rope pumps, handpumps, India Mark III, wind-powered, and motorpumps are relatively rare.

Insight:

- Water extraction technologies in Tanzania reflect a mix of traditional and mechanized approaches.

Bivariate Analysis

1. How does extraction type affect water well status?



The figure above represents the relationship of water well status by the technology used to extract water from it

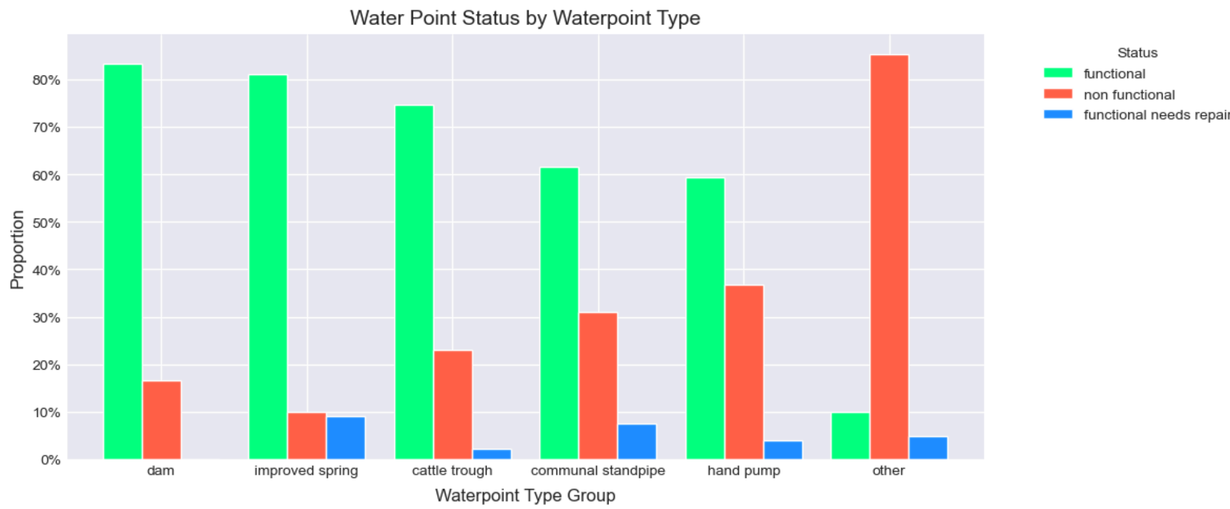
Observation:

- Water point functionality varies significantly by extraction technology.
- Motorpumps (88.9%) and rope pumps (78.4%) have the highest proportion of fully functional points, while traditional or less common technologies like India Mark III (27.0%), wind-powered (27.5%), and “other” types (13.4%) show the lowest functionality and the highest non-functional rates.
- Gravity systems, which are the most widely used, have a fairly high functionality rate (63.2%) but also a notable share of points needing repair (9.0%).
- Handpumps, mono, and submersible systems display moderate functionality but substantial non-functional percentages, highlighting maintenance challenges.

Insight:

- Overall, technology choice strongly influences water point reliability, with modern and mechanically robust systems generally performing better than older or less standardized types.

2. Are certain waterpoint types more prone to failure?



The figure above represents the relationship between water point status and water point type

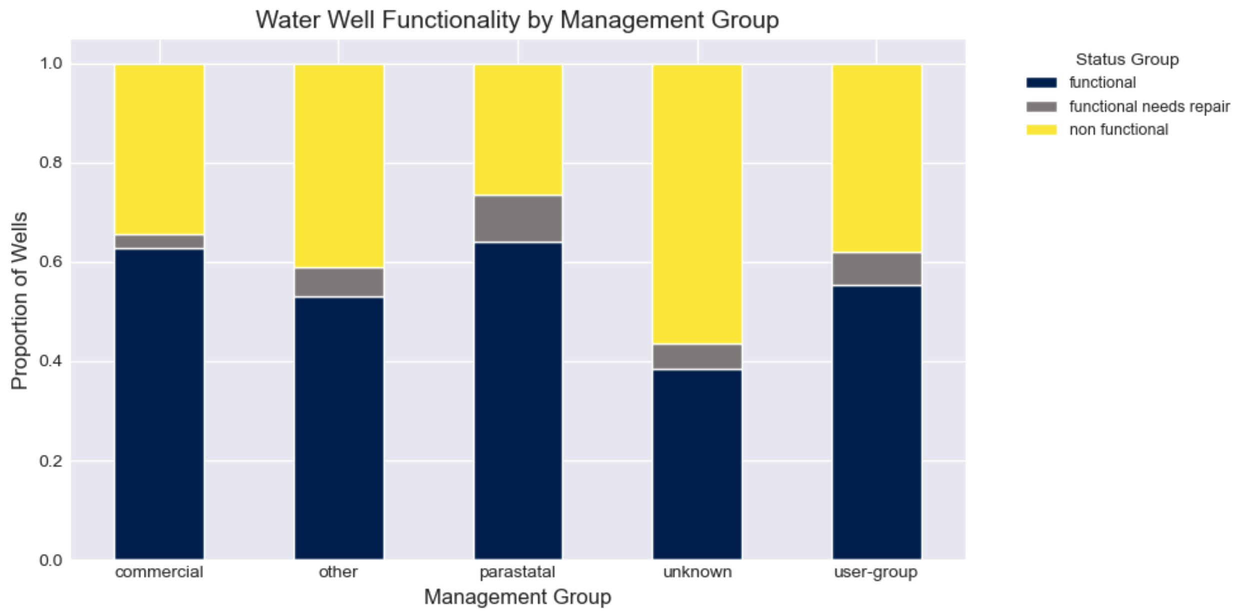
Observation:

- Water point functionality varies by type.
- Dams show perfect functionality (100%), while improved springs (81.0%) and cattle troughs (80.6%) also perform well.
- Communal standpipes (61.5%) and hand pumps (59.2%) have moderate functionality, with a significant share being non-functional, highlighting maintenance needs in commonly used community infrastructure.
- Points classified as "other" perform poorly, with only 9.9% functional and 84.9% non-functional.

Insight:

- Overall, specialized or well-managed water points tend to be more reliable, whereas widely used community types face higher failure rates.

3. Does management influence the likelihood of a well being functional?



The figure above shows how water well functionality differs depending on who manages them

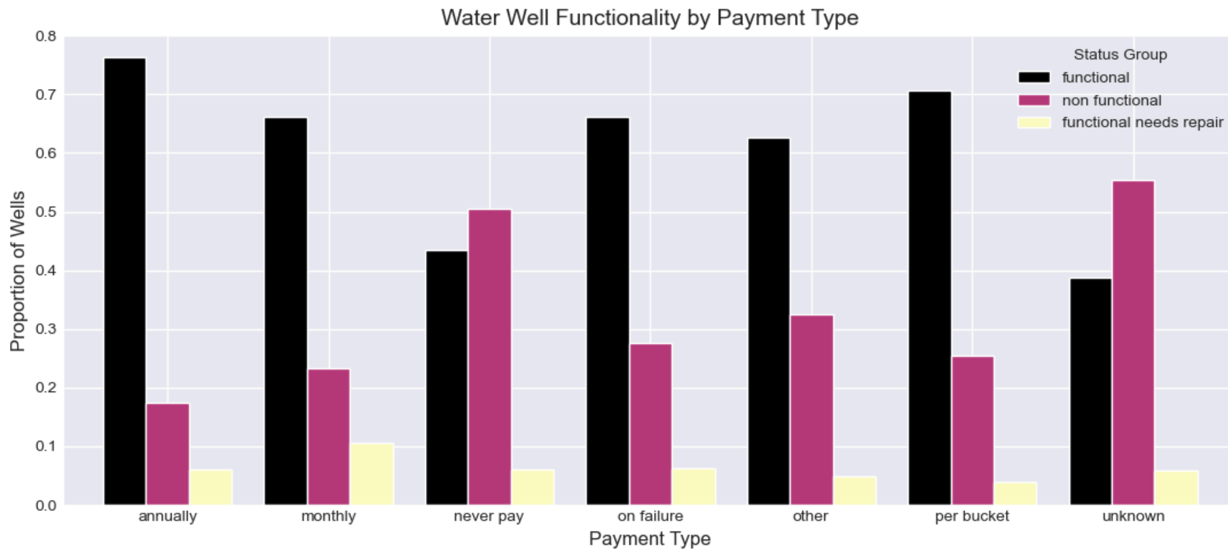
Observation:

- Water point functionality varies by management type.
- Parastatals show the highest functionality (63.4%) with a relatively low non-functional rate (26.8%), suggesting effective oversight.
- Commercial (58.9%) and user-group managed points (56.0%) perform moderately, while points under “other” management types also show similar patterns.
- Water points with unknown management perform the worst, with only 37.3% functional and 57.4% non-functional, highlighting the risks of unclear oversight.

Insight:

- Effective management appears strongly linked to higher water point reliability.

4. How does payment relate to well functionality?



The figure above represents the relationship between water well functionality and payment

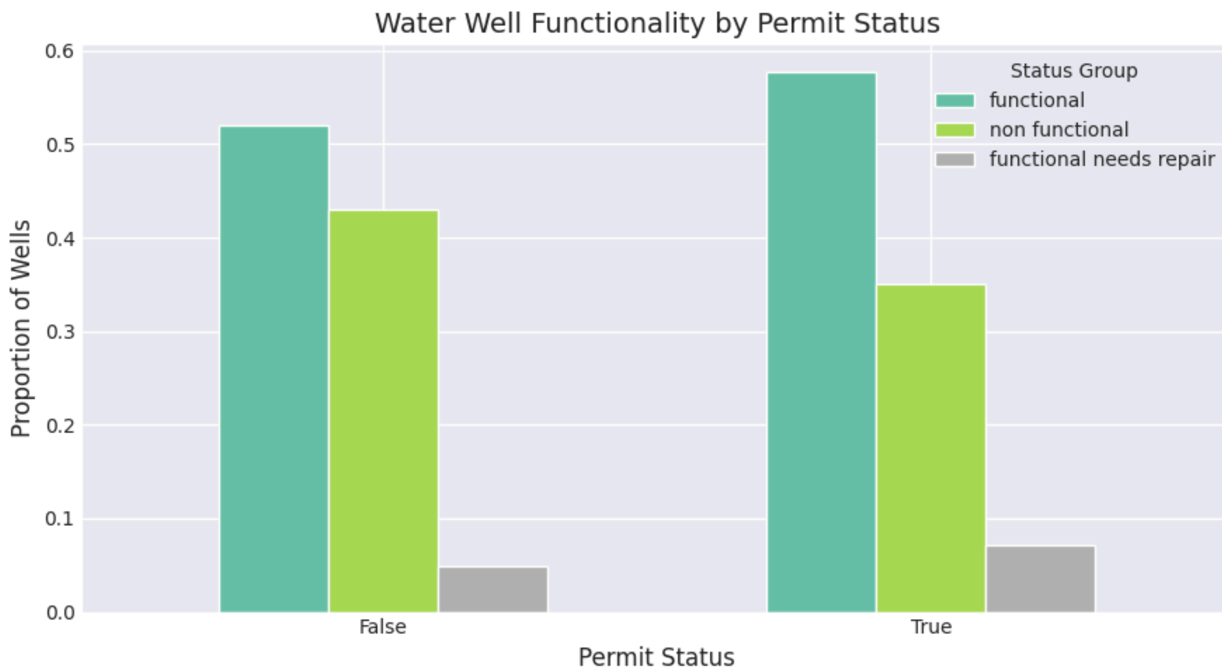
Observation:

- Water point functionality is closely linked to payment type.
- Points with regular payments; such as annual (76.3%) or per bucket (69.7%) contributions, tend to be more functional, while those in the “never pay” (42.7%) or unknown (42.3%) categories show the lowest functionality and the highest non-functional rates.
- Monthly payments and “on failure” schemes perform moderately, with functionality around 66–67%.

Insight:

- Consistent user contributions appear to support better water point reliability, whereas non-payment or unknown payment arrangements correlate with higher failure rates.

5. Do wells with permits perform better than those that don't have permits?



The figure above shows the status of water wells in relation to their legal compliance

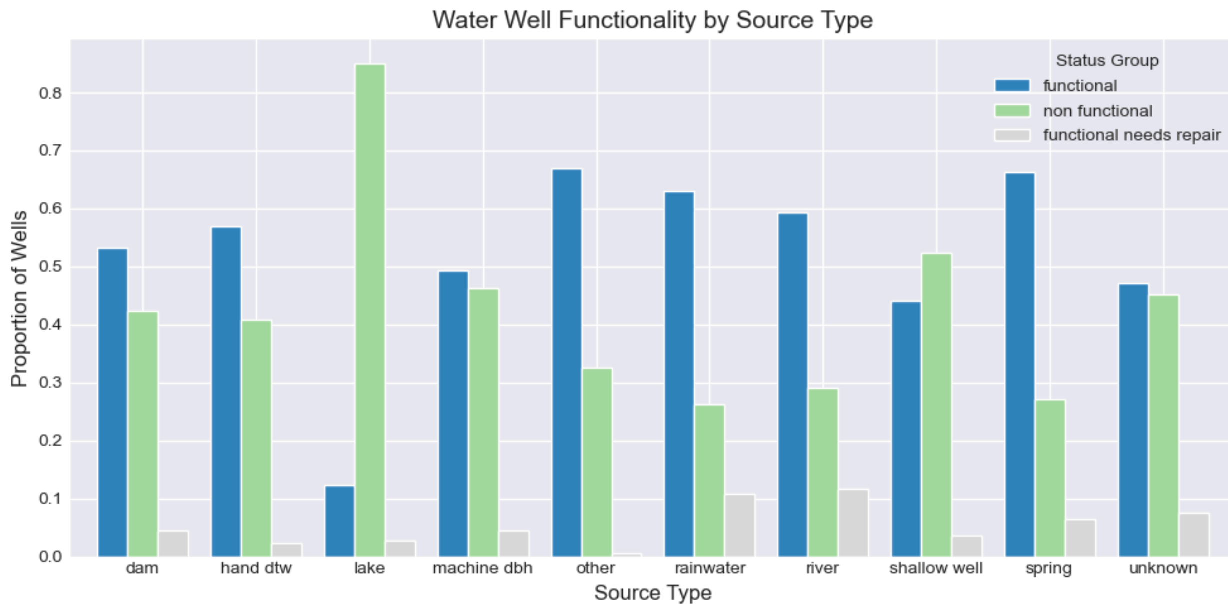
Observation:

- Water points with permits generally perform better, with 57.8% functional compared to 52.0% for those without permits.
- Non-permitted points have a higher non-functional rate (43.0% vs. 35.0%).
- Points needing repairs are slightly more common among permitted water points (7.2% vs. 5.0%).

Insight:

- Having a permit appears associated with slightly higher reliability, suggesting that formal authorization may contribute to better maintenance and oversight.

6. Are certain water sources more reliable than others?



The figure above shows the operational status of wells with different sources

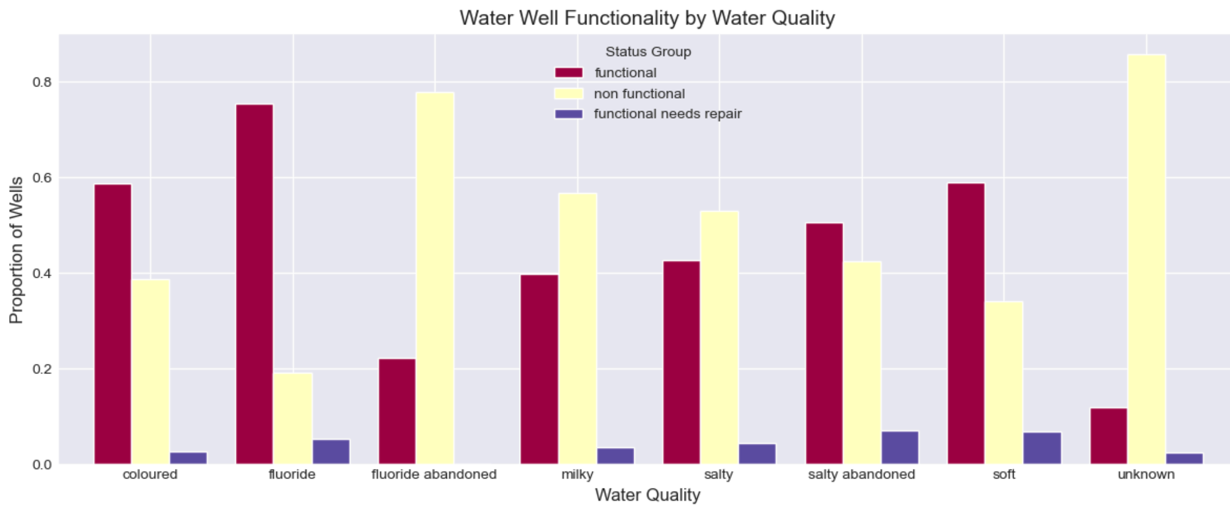
Observation:

- Water point functionality varies widely by source.
- Springs (66.4%) and “other” sources (68.0%) show the highest functionality, while lakes have the lowest, with only 12.3% functional and 85.0% non-functional.
- Rivers (59.4%) and rainwater (61.0%) systems perform moderately well.
- Shallow wells (44.8%) and machine-drilled boreholes (47.9%) have nearly equal shares of functional and non-functional points, highlighting maintenance challenges.

Insight:

- Natural springs and well-managed specialized sources are the most reliable, whereas surface water sources like lakes face significant operational challenges.

7. Does water quality correlate with functionality status?



The figure above represents the relationship between water quality and well functionality

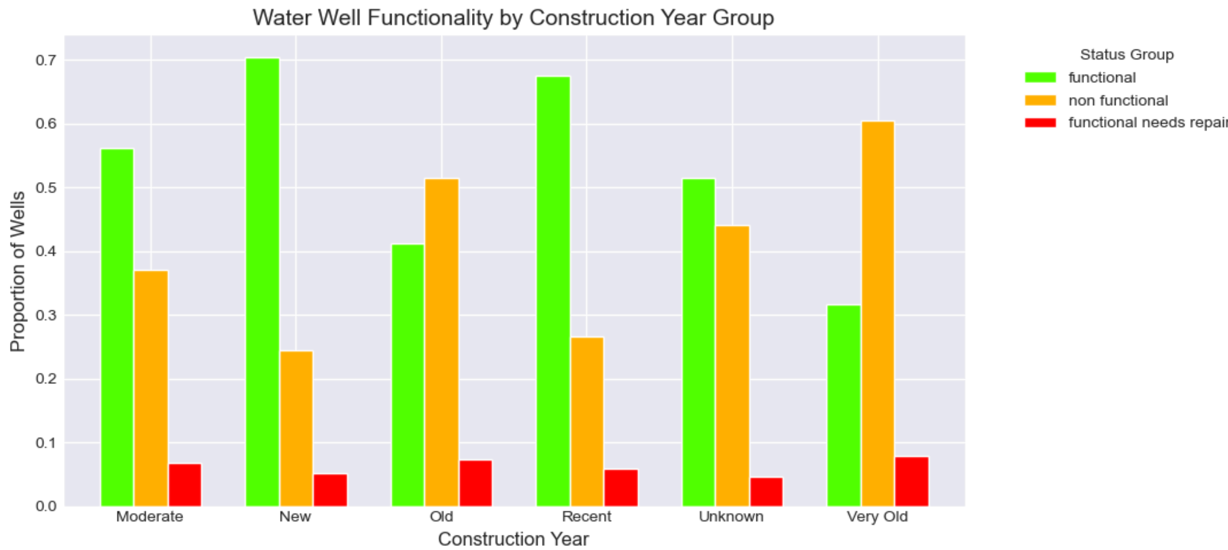
Observation:

- Water point functionality varies considerably by water quality.
- Points with fluoride (75.2%) and coloured water (58.8%) are among the most functional, while soft water points also perform reasonably well (58.5%).
- Poor-quality sources; milky (38.7%), salty (42.3%), salty abandoned (33.8%), and fluoride abandoned (28.6%), show high non-functional rates, with unknown-quality points performing the worst (15.4% functional, 82.7% non-functional).

Insight:

- Water quality is strongly linked to reliability, with contaminated or abandoned sources facing the greatest operational challenges.

8. Do older wells tend to fail more than newer ones?



The figure above shows the status of wells by the year they were constructed.

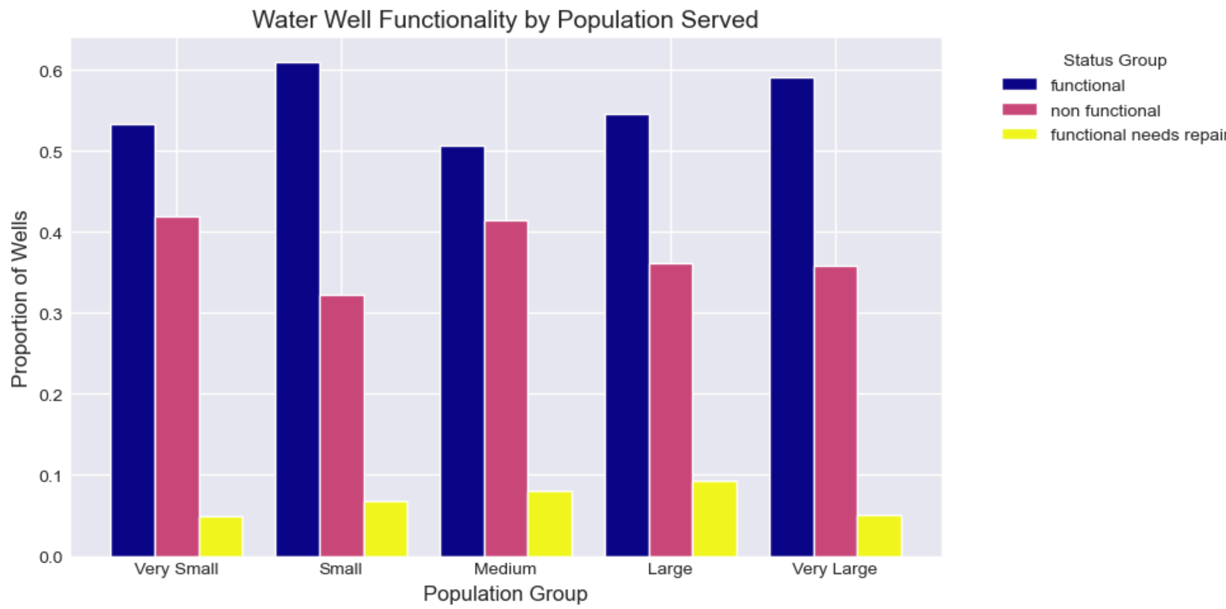
Observation:

- Functionality strongly depends on the age of the water point.
- New (70.3%) and recent (67.2%) constructions are the most reliable, while old (41.9%) and very old (32.6%) points have the highest non-functional rates.
- Moderately aged points perform moderately well (56.4%), and points with unknown construction year show mixed outcomes.

Insight:

- Newer infrastructure is clearly associated with higher reliability.

9. Does the population a well serves affect its functionality?

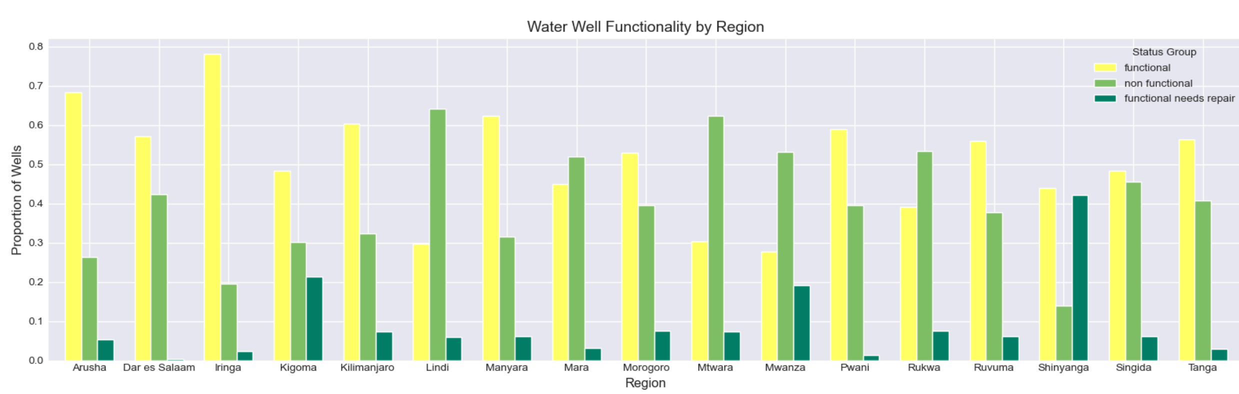


The figure above shows how operational a well is by the number of people it serves

Observation:

- Water point functionality varies modestly by the size of the population served.
- Small (60.7%) and very large (58.5%) water points are the most functional, while medium-sized points have lower functionality (49.8%).
- Very small (55.3%) and large (53.7%) points are moderately functional.
- Non-functional rates are highest among medium-sized points (42.0%), suggesting that population demand may influence maintenance and performance.

10.Regional differences on well functionality by region.



The figure above represents the origin of water at each point

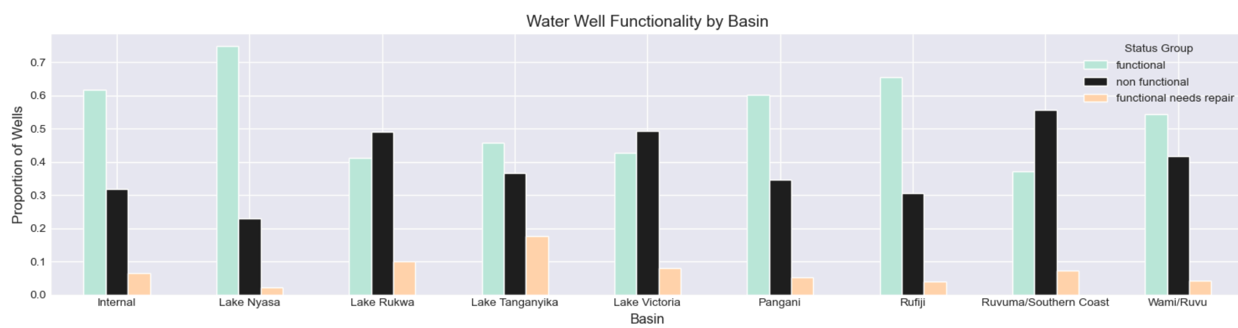
Observation:

- Functionality varies considerably across regions.
- Iringa (78.2%) and Arusha (69.7%) show the highest proportion of functional points, while regions like Mwanza (27.8%), Lindi (29.8%), and Mtwara (30.3%) perform poorly.
- Some regions, such as Kigoma and Shinyanga, have unusually high rates of points needing repair, indicating localized infrastructure or maintenance challenges.

Insights:

- There is strong regional disparity in water point reliability.

11.Regional differences on well functionality by basin.



The figure above represents the origin of water at each point

Observation:

- Lake Nyasa (74.8%) and Internal basins (67.4%) have the highest functional rates, while basins like Ruvuma/Southern Coast (37.2%) and Lake Victoria (40.4%) show high non-functional percentages.
- Other basins, such as Lake Rukwa (41.1%) and Lake Tanganyika (45.7%), also display moderate functionality with notable repair needs.
- This suggests that basin-specific conditions, resource management, or maintenance practices strongly influence water point reliability.

Conclusions and Recommendations

Tanzania’s water points are mostly functional, but many are non-functional or in need of repair. Community-managed systems dominate, yet reliability varies with technology, water quality, and maintenance; gravity-fed, rope, and motorpumps outperform older or less standardized types, while salty, milky, or abandoned sources fail more often. Regional and basin disparities are pronounced, with Iringa, Arusha, Lake Nyasa, and Internal performing best, and Mwanza, Lindi, Mtwara, Lake Victoria, and Ruvuma/Southern Coast lagging. Formal permits and consistent payment schemes improve reliability, highlighting the need for targeted interventions to strengthen access, maintenance, and sustainability.

Conclusions

1. **Functionality Variability:** While most water points work, a substantial portion is non-functional or requires repair, particularly older infrastructure and less standardized technologies.
2. **Management Impact:** Community-managed points dominate but require technical support and capacity building to maintain reliability.
3. **Technology and Water Quality:** Modern extraction technologies and higher-quality water sources correlate strongly with functionality; poor-quality or abandoned sources perform worst.
4. **Regional Disparities:** Significant differences exist across regions and basins, highlighting areas needing prioritized interventions.
5. **Population and Payment Influence:** Medium-sized points and non-paying or unpermitted water points show higher failure rates, emphasizing the importance of sustainable funding and formal oversight.

Recommendations

1. **Targeted Maintenance:** Prioritize repairs for older infrastructure, non-functional points, and regions/basins with high failure rates.
2. **Strengthen Community Management:** Train and support Village Water Committees and user-groups to improve maintenance and oversight.
3. **Promote Reliable Payment Systems:** Encourage regular payment schemes (annual or per-bucket) to fund maintenance and sustainability.
4. **Invest in Proven Technologies:** Favor gravity-fed systems, motorpumps, and rope pumps, while phasing out or upgrading high-failure technologies.
5. **Monitor and Improve Water Quality:** Regular testing and mitigation in areas with poor-quality or abandoned water points.
6. **Data and Planning:** Fill gaps in unknown water points’ management, payment, or quality to inform resource allocation and strategic planning.

Modeling

Target Definition & Preprocessing

During preprocessing, the “functional needs repair” class was dropped from the target variable.

Justification:

- This class represented only 6.58% of the data, making it statistically insignificant and highly imbalanced.
- Operationally, the label is ambiguous. Wells needing repair often transition into either functional or non-functional categories in reality.
- Keeping it introduced noise and instability into model learning with little gain in actionable insights.

By dropping this class, the problem was reframed as a binary classification task:

1. Functional wells
2. Non-functional wells (treated as the minority and the class of highest interest).

This reframing improved model performance and interpretability, though it reduced granularity on borderline repairable wells.

Approach

Several algorithms were trained and tuned:

- Baseline models: Logistic Regression, Decision Tree, KNN.
- Ensemble models (main focus): Random Forest, XGBoost, and CatBoost.

Three strategies were applied:

1. Vanilla training.
2. Class weighting to counter class imbalance.
3. SMOTE (Synthetic Minority Oversampling Technique) to resample the training data.

For ensemble models, hyperparameter optimization was performed using both GridSearchCV and RandomizedSearchCV with 5-fold cross-validation.

Evaluation Metrics

The models were evaluated using:

- Test accuracy (overall correctness).

- Recall for non-functional wells (critical to detect failures).
- Precision and F1-score (balancing false positives and false negatives).
- Confusion matrices for intuitive error breakdown.
- ROC-AUC Curve

Results (Tuned Models)

- **Random Forest with SMOTE and RandomizedSearchCV:**
    - Training accuracy: 80.31%
    - Test accuracy: 80.30%
    - Minority recall: 82%.
    - Precision (Macro): 0.79
    - Recall (Macro): 0.80
    - F1-score (Macro): 0.79

  Notes: Balanced performance, robust handling of the minority class.

- **Random Forest with SMOTE and GridSearchCV:**
    - Training accuracy: 100.00%
    - Test accuracy: 86.01%
    - Minority recall: 82%.
    - Precision (Macro): 0.85
    - Recall (Macro): 0.86
    - F1-score (Macro): 0.85

  Notes: Strong performance but signs of overfitting due to perfect training accuracy.

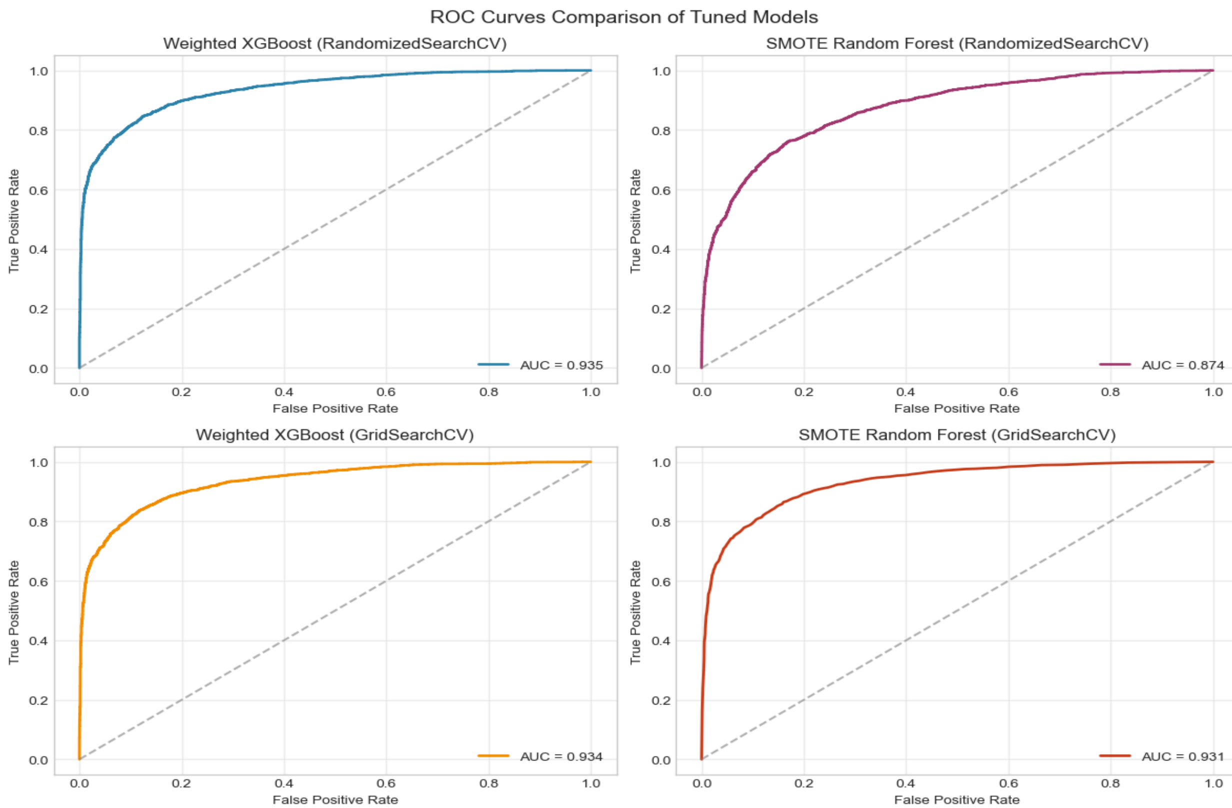
- **XGBoost with class weights and GridSearchCV:**
    - Training accuracy: 94.56%
    - Test accuracy: 86.77%
    - Minority recall: 82%.
    - Precision (Macro): 0.86
    - Recall (Macro): 0.87
    - F1-score (Macro): 0.86

  Notes: Best performing model overall with strong accuracy and stable training.

- **XGBoost with class weights and RandomizedSearchCV:**
    - Training accuracy: 96.14%
    - Test accuracy: 86.48%
    - Minority recall: 82%.
    - Precision (Macro): 0.86
    - Recall (Macro): 0.86
    - F1-score (Macro): 0.86

Notes: Very strong, but marginally behind the GridSearchCV variant.

ROC-AUC CURVE Analysis



- The ROC curves provide a comparative view of the discriminative power across the tuned models.
- Weighted XGBoost consistently demonstrates the strongest performance, with AUC scores of 0.935 (RandomizedSearchCV) and 0.934 (GridSearchCV), reflecting both high accuracy and stability across tuning approaches.
- In contrast, the SMOTE Random Forest shows a larger gap between search strategies: the RandomizedSearchCV variant achieves 0.874, while GridSearchCV boosts performance to 0.931, nearly closing the gap with XGBoost.
- This highlights Random Forest's sensitivity to hyperparameter optimization, as thorough grid search substantially improves results.
- Overall, the analysis indicated that Weighted XGBoost remains the most reliable and highest-performing model, while SMOTE Random Forest, when carefully tuned, offers competitive results.
- The close AUC values for XGBoost under both search methods also underscore its robustness, whereas Random Forest benefits more noticeably from exhaustive tuning.

Insights

- All tuned models consistently identified about 82% of non-functional wells, a significant improvement given the initial imbalance.
- The top-performing model was XGBoost with class weights and GridSearchCV, which achieved the highest test accuracy (86.77%) without extreme overfitting.
- Random Forest with SMOTE and GridSearchCV was competitive but showed signs of overfitting.
- XGBoost weighted models are faster and more efficient to deploy than Random Forest trained with SMOTE.

Model Limitations

- **Data drift and timeliness:** The dataset reflects historical conditions. Management practices, technology usage, and environmental stress may have changed since collection, so predictions on today's wells may be less accurate without periodic retraining.
- **Dropped class granularity:** The "functional but needs repair" category was excluded to stabilize and simplify the model. This improves performance but loses detail about borderline cases. In practice, some wells flagged as functional may still need attention.
- **Regional and basin heterogeneity:** Model performance can vary by region, basin, or extraction technology. A model trained on national data may underperform in areas with unique conditions or very few records.
- **Incomplete and inconsistent data:** Unknown categories can lead to biased predictions for underrepresented groups.
- **Non-static factors:** Sudden shocks (droughts, floods, contamination) and behavioral changes aren't captured by static historical features, so the model may miss emerging risks.
- **Operational deployment risks:** If predictions are used directly to prioritize maintenance, false negatives (failing wells missed) could leave communities without water; false positives (healthy wells flagged) could waste scarce maintenance resources. A human-in-the-loop review is recommended.

Recommendations

1. Adopt XGBoost with class weights and GridSearchCV as the final model for deployment.
2. Validate the model on the untouched holdout test dataset for confirmation.
3. Continue monitoring Random Forest models as backups, but treat them cautiously due to overfitting risk.
4. Should predictions for "functional needs repair" be required, train a dedicated model or collect more data for that class.
5. Monitor the deployed model's recall on non-functional wells and retrain periodically with updated data.

Conclusion:

The modeling pipeline produced strong, stable classifiers. By reframing the problem as binary and applying ensemble learning with proper class imbalance handling, the final solution balances accuracy, robustness, and operational usefulness for detecting non-functional wells.